Project Report
On
<u>CUSTOMER RELATIONSHIP MANAGEMENT SYSTEM</u>

Submitted to

LOVELY PROFESSIONAL UNIVERSITY

In partial fulfillment of the requirements for the award of degree of

Master of Computer Applications

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CERTIFICATE

This is to certify that Subha Sankar Chakraborty (11402181) has completed doing full term industrial internship titled “Customer Relationship Management” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original study. No part of the report has ever been submitted for any other degree or diploma.

The report is fit for the submission and the partial fulfillment the conditions for the award of Master of Computer Application.

Date:

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We also extend my heartfelt thanks to my family and well-wishers who supported mentally for completing this thesis.

Lastly, we offer my regards to all of those who supported me in any respect during the completion of the project.

Date: ……..…

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CHAPTERS

CHAPTER 1-INTRODUCTION

1.1. Introduction to project

A CRM is a collection of people, processes, software, and internet capabilities that helps an enterprise manage customer relationship effectively and systematically. The goal of CRM is to understand and anticipate the needs of current and potential customer to increase customer retention and loyalty while optimizing the way product and services are sold.

CRM stands for Customer Relationship Management. It is a strategy used to learn more about customers' needs and behaviors in order to develop stronger relationships with them. After all, good customer relationships are at the heart of business success. There are many technological components to CRM, but thinking about CRM in primarily technological terms is a mistake. The more useful way to think about CRM is as a process that will help bring together lots of pieces of information about customers, sales, marketing effectiveness, responsiveness and market trends

The objective is to capture data about every contact a company a company has with a customer through every channel and store it in the CRM system to enable the company to truly understand customer action. CRM software helps an organization build a database about its customer that management, sales people, customer service provider and even customer can access information to access customer needs with product and offering.

Marketing Automation is the most comprehensive campaign management solution available. It provides everything needed to turn raw, disparate customer data into profitable marketing campaigns – all the way through inception, execution and measurement.

Marketing Optimization applies sophisticated mathematical approaches to optimize marketing campaign ROI given limited budgets, channel capacities and other organizational constraints.

The idea of CRM is that it helps businesses use technology and human resources to gain insight into the behavior of customers and the value of those customers. If it works as hoped, a business can:

- Provide better customer service
- Make call centers more efficient

- Cross sell products more effectively
- Help sales staff close deals faster
- Simplify marketing and sales processes
- Discover new customers
- Increase customer revenues

1.2 ARCHITECTURE OF CRM

There are three parts of application architecture of CRM:

1. Operational - automation to the basic business processes (marketing, sales, service)
2. Analytical - support to analyze customer behavior, implements business intelligence alike technology
3. Collaborative - ensures the contact with customers (phone, email, fax, web, sms, post, in person)

Operational CRM

Operational CRM means supporting the so-called " front office" business processes, which include customer contact (sales, marketing and service). Tasks resulting from these processes are forwarded to employees responsible for them, as well as the information necessary for carrying out the tasks and interfaces to back-end applications are being provided and activities with customers are being documented for further reference.

According to Gartner Group, the operational part of CRM typically involves three general areas of business:

- **Sales Force Automation (SFA):** SFA automates some of the company's critical sales and sales force management functions, for example, lead/account management, contact management, quote management, forecasting, sales administration, keeping track of customer preferences, buying habits, and demographics, as well as sales staff performance. SFA tools are designed to improve field sales productivity. Key infrastructure requirements of SFA are mobile synchronization and integrated product configuration.
- **Customer Service and Support (CSS)**: CSS automates some service requests, complaints, product returns, and information requests. Traditional internal help desk and traditional inbound call-center support for customer

inquiries are now evolved into the "customer interaction center" (CIC), using multiple channels (Web, phone/fax, face-to-face, kiosk, etc). Key infrastructure requirements of CSS include computer telephony integration (CTI) which provides high volume processing capability, and reliability.

- **Enterprise Marketing Automation (EMA)**: EMA provides information about the business environment, including competitors, industry trends, and macroenviromental variables. It is the execution side of campaign and lead management. The intent of EMA applications is to improve marketing campaignefficiencies. Functions include demographic analysis, variable segmentation, and predictive modeling occur on the analytical (Business Intelligence) side.

Analytical CRM

In analytical CRM, data gathered within operational CRM and/or other sources are analyzed to segment customers or to identify potential to enhance client relationship. Analysis of Customer data may relate to one or more of the following analyses:

1. Contact channel optimization
2. Contact Optimization
3. Customer Segmentation
4. Customer Satisfaction Measurement / Increase
5. Sales Coverage Optimization
6. Pricing Optimization
7. Product Development
8. Program Evaluation

Data collection and analysis is viewed as a continuing and iterative process. Ideally, business decisions are refined over time, based on feedback from earlier analysis and decisions. Therefore, most successful analytical CRM projects take advantage of a data warehouse to provide suitable data.

Collaborative CRM

Collaborative CRM facilitates interactions with customers through all channels (personal, letter, fax, phone, web, e-mail) and supports co-ordination of employee teams and channels. It is a solution that brings people; processes and data together so companies can better serve and retain their customers.

Collaborative CRM provides the following benefits:

- Enables efficient productive customer interactions across all communications channels
- Enables web collaboration to reduce customer service costs
- Integrates call centers enabling multi-channel personal customer interaction
- Integrates view of the customer while interaction at the transaction level

1.3 BENEFITS OF CRM SYSTEM

Give customer-facing employees the best tools available

Give your sales and service professionals the chance to deliver stellar customer service every time they are in a call or at a customer site. With optimally configured software solutions from Microsoft Business Solutions, they can find information quickly, answer customer questions satisfactorily, and ensure that your business fulfills or exceeds customer commitments.

Deliver high-quality service at all times

Happy customers generate return business, referrals, and excellent visibility for your business. You want to make doing business with your company efficient and rewarding for them—track customer preferences, be aware of their account histories, and resolve any issues promptly. Microsoft Business Solutions can empower your people to streamline customer communications and interactions, offer timely, reliable information, and demonstrate your organization's accountability.

Boost sales and service performance

Automation of business tasks can go a long way towards streamlining sales and customer service processes, shortening sales cycles, and maintaining consistent quality and productivity on your sales and service teams. In addition, Microsoft Business Solutions offerings allow you to review and enhance the performance of individual sales and service employees as well as entire teams, forecast and plan sales activity, and act on new business opportunities

1.4 OBJECTIVE OF PROJECT

Objectives of the projects are as following-

i. To manage customer data effectively

ii. To help the sales people in working easily and efficiently.
iii. To manage sales person data effectively
iv. To evaluate sales performance of different sales person.
v. To generate sales report
vi. To perform analysis on customer data
vii. Simplify marketing and sales processes
viii. To help the sales people in reducing errors that are encountered frequently during manual operations by concurrently updating the data stored in many places.
ix. To provide security as only an authorized user can interact with the system
x. To Provide product information to the customer
xi. Provide a fast mechanism for correcting service deficiencies.
xii. To provide mechanism to deal with customer complaints effectively.
xiii. To provide list of customers whenever needed by salesman, administrator or customer relationship executive.
xiv. To send Offer Letter to customers by finding right customer group.
xv. To do customer segmentation on basis of income, occupation and territory.
xvi. To perform analysis of sales data on the basis of product, salesman, and territory.
xvii. To assign quota of salesman.
xviii. To assign territory to salesman.
xix. To provide facility of submitting feedback of the customer.

CHAPTER 2: SYSTEM ANALYSIS

2.1 Introduction to System Analysis

For identifying the need of the System, we consider the following:

The performance of the System

The performance of the System basically depends on the System Design, tools used in the system, coding, etc. The performance of the system should be fast, accurate & reliable.

The information being supplied and its form

The information being supplied should be very precise, correct and updated. All the information regarding project should be collected from the reliable source. In Project “Customer Relationship Management”, all the information regarding

- Customer
- Product
- Transactions
- Complaint
- Offering
- Schedule
- Customer feedbacks

Should be collected from the Management & Sales People of the organization.

The existing system in the organization is not completely computerized. The system is not working smoothly; therefore the organization has decided to replace it with a completely computerized one. The problems, which the existing system faces, are:

2.2 Problem with Existing System

Low Functionality:

With the existing system, the biggest problem was the low functionality of the department. The problem faced hampered the work of the department. For all the tasks like entering the customer data, salesman data, product data, taking the orders, making Bill, making reports etc a large number of employees were appointed who would have been utilized in some other useful tasks.

Erroneous Input And Output:

In the existing system, humans performed all tasks. As in the human tendency, errors are also a possibility. Therefore, the inputs entered by the salesman in the registers may not be absolutely foolproof and may be erroneous. As a result of wrong input, the output reports etc will also be wrong which would in turn affect the performance of home appliance company.

Portability:

System that existed previously was manual. As a result, the system was less portable. One has to carry the loads of so many registers to take the data from one place to another. A big problem was that the system was less flexible and if one wanted to make a change would need to change in all the registers that would also prove to be big headache.

Security:

Security concerns were also one of the motives of the department for the need of the software. In the registers, the data is not secure as anybody can tamper with the data written in the registers. Also for the security of the registers, lots of problems arise to store the registers in a secure place and the appointments of security personnel can also cost a bit. Whereas in the software, just a password makes it absolutely secure from the reach of unauthorized users.

Data Redundancy:

In the case of manual system, the registers are maintained in which, a lot of data is written. Therefore, there is a problem in the registers that the same data may be repeated again and again. Against the customer id, a lot of data will be repeated which will cause a lot of problems at the time of query as well as at the time of preparing the reports because a single data that will be left mistakenly will largely affect the report and subsequently, the performance of the department. In the software the concept of primary key and foreign key is used very efficiently, which will prevent the redundancy of data that will prove to be very beneficial to the organization because it will nullify the human error completely.

Processing Speed:

In manual system, for a simple work, a number of employees are appointed and in case of keeping records of orders given by the customers, making Bill & reports, they take a lot of time, which may affect the performance of the organization as well as hamper the progress of the organization. It also affects the speed of working in the organization and the work that should have been performed in very short duration can take a large amount of time. But, in the case of software, all the tasks are performed at the touch of a key, which improves the performance of the organization, a great deal, thereby, improving the chances of progress of the organization.

Manual Errors:

When a number of tough tasks are prepared by the humans like preparation of reports, keeping records of all the customers, salesman & available stock in company etc then some human errors are obvious due to a number of factors like mental strain, tiredness etc. But as we all know that computer never gets tired irrespective of the amount of work it has to do. So, this software can nullify the probability of manual errors that improves the company performance.

Complexity in Work:

In a manual system, whenever a record is to be updated or to be deleted a lot of cutting and overwriting needs to be done on all the registers that are concerned with the deleted or updated record, which makes the work very complex. However in the software, once a record is updated or deleted, all the concerned changes are made in the database automatically.

The efficiency of the existing System

The existing System is not well efficient which provokes to develop the new System with some modifications in the older system wherever required. If the existing System is not that much efficient, as required for better results & performance, then it should be converted into the new advanced & more efficient System. The new System should be well efficient, fast as compared with the older one to show better results than previous one.

The security of the data and software

The Security of both Data & Software is the most important and prime thing in the System Analysis & Design. Data & Software should be kept secured from the unreliable sources otherwise it leads to the Piracy of both data & Software. For proper security username and password is provided to every user of the system. They can access the system as per their authority.

The security of the equipment and personnel, etc

The security of the equipment and personnel, etc is also very important in the System design. Proper measures are to be taken for the security of Equipments, Personnel, etc for the proper functioning of the System.

2.3 PRIMARY INVESTIGATION

Evaluation of project request is major purpose of preliminary investigation. It is the collecting information that helps committee members to evaluate merits of the project request and make judgment about the feasibility of the proposed project.

At the heart of any system analysis is detailed understanding of all-important facts of the business area under investigation. The key questions are-

- What is being done?
- How it is being done.
- How frequently does it occur?
- How great is the volume of transactions or decision.
- How well is the task being performed?
- Does a problem exist?
- If a problem exists, how serious is it? What is the underlying cause?

To answer the above questions, system analysts discuss with different category of person to collect facts about their business and their operations.

When the request is made, the first activity the preliminary investigation begins. Preliminary investigation has three parts-

- **Request clarification**
- **Feasibility study**
- **Request approval**

Request Clarification

An information system is intended to meet needs of an organization. Thus the first step is in this phase is to specify these needs and requirements. The next step is to determine the requirements met by the system. Many requests from employees and users in the organizations are not clearly defined. Therefore, it become necessary that project request must examine and clarified properly before considering system investigation.

Information related to different needs of the System can be obtained by different users of the system. This can be done by reviewing different organization's documents such as current method of storing sales data, complaint data etc. By observing the onsite activities the analyst can get close informations related to real system.

2.4 FEASIBILITY STUDY

A feasibility study is a preliminary study undertaken before the real work of a project starts to ascertain the likelihood of the project's success. It is an analysis of all possible solutions to a problem and a recommendation on the best solution to use. It involves evaluating how the solution will fit into the corporation. It, for example, can decide whether an order processing be carried out by a new system more efficiently than the previous one.

2.4.1 **Technical Feasibility**

Technical feasibility refers to the ability of the process to take advantage of the current state of the technology in pursuing further improvement. The technical capability of the personnel as well as the capability of the available technology should be considered. Number of technical issues which are generally realized during the feasibility stage of investigations-

- Present technology, which is in case of this system is manual data management which is highly inefficient and inconsistent, o Technical capacity of proposed equipments to hold data required using the system. Upgradability of system if developed

o Technical guarantee of accuracy, reliability, ease of access and data security.

2.4.2 Economic feasibility

one of the most frequently used technique for evaluating effectiveness of the system is economic analysis. The procedure is to determine the benefits and saving that are expected with the proposed system and compare it with cost. This involves the feasibility of the proposed project to generate economic benefits. The analyst can conduct analysis of economic feasibility by considering following issues-

o The cost to conduct a full systems investigation.

o The cost of hardware and software for class of application being consider. o The benefits in the form of reduced cost or fewer costly errors.

In case of “Customer Relationship Management” the company has to maintain large amount of manpower, money and time resources. Huge amount of money can be saved if this system is automated.

2.4.3 Operational Feasibility

This feasibility is related to human organizational and political aspects. This feasibility is carried out by a small group of people who are familiar with information system technique and
understand the parts of the business that are relevant to the project and are skilled in system analysis and design process**.** Issues involve in testing operational feasibility-

o Support of management

o Acceptability of current business method o Need of change in existing system

o Involvement of user in planning and development of the project o Effect of implementation of the proposed system.

Request Approval

All projects that are submitted for evaluation are not feasible. In general, requests that do not pass all feasibility tests are not perused further unless they are modified and resubmitted as new proposal. In some cases, it also happens that a part of a newly developed system is unworkable and the selection committee may decide to combine the workable part of the project with another feasible proposal.

The project development is feasible technically, operationally and economically so it is feasible to develop the project.

2.5 SOFTWARE ENGINEERING PARADIGM APPLIED

Software Engineering is the establishment and use of sound engineering principles in order to obtain economically software that is both reliable and works efficiently on real systems.

To solve actual problems in an industry / organizations setting, such as Central or State Govt. Sectors, Public / private Sectors, Colleges, Schools, etc., a Software Engineer or a team of engineers must incorporate a development strategy that encompasses the process, methods, tools layers, and the generic phases. This strategy is referred to as a Process Model or a Software Engineering Paradigm.

Models used for system development in project "Customer Relationship Management", is Prototyping. Descriptions of these models are as following-

PROTOTYPE MODEL

Prototyping is the process of building a model of a system. In terms of an information system, prototypes are employed to help system designers build an information system that intuitive and easy to manipulate for end users. Prototyping is an iterative process that is part of the analysis phase of the systems development life cycle.

During the requirements determination portion of the systems analysis phase, system analysts gather information about the organization's current procedures and business processes related the proposed information system. In addition, they study the current information system, if there is one, and conduct user interviews and collect documentation. This helps the analysts develop an initial set of system requirements.

Prototyping can augment this process because it converts these basic, yet sometimes intangible, specifications into a tangible but limited working model of the desired information system. The user feedback gained from developing a physical system that the users can touch and see facilitates an evaluative response that the analyst can employ to modify existing requirements as well as developing new ones.

The prototyping model is a software development process that begins with requirements collection, followed by prototyping and user evaluation. Often the end users may not be able to provide a complete set of application objectives, detailed input, processing, or output requirements in the initial stage. After the user evaluation, another prototype will be built based on feedback from users, and again the cycle returns to customer evaluation. The cycle starts by listening to the user, followed by building or revising a mock-up, and letting the user test the mock-up, then back.

Some Advantages of Prototyping:

- Reduces development time.
- Reduces development costs.
- Requires user involvement.
- Developers receive quantifiable user feedback.
- Facilitates system implementation since users know what to expect.
- Results in higher user satisfaction.
- Exposes developers to potential future system enhancements.

2.6 TOOLS AND TECHNOLOGY USED

- JAVA SE DEVELOPMENT KIT (JDK)
- ANDROID SDK
- SQL LITE
- ANDROID STUDIO

JDK

The Java Development Kit (JDK) is an execution of both of the Java SE, Java EE or Java ME stages expel by Oracle Corporation as a parallel item went for Java designers on Solaris, Linux, Mac OS X or Windows. The JDK incorporates a private JVM and a couple of different assets to complete the formula to a Java Application. Since the presentation of the Java stage, it has been by a wide margin the most generally utilized Software Development Kit (SDK).

A Java Development Kit (JDK) is a project improvement environment for composing Java applets and applications. It comprises of a runtime situation that “sits on top” of the working framework layer and also the apparatuses and programming that engineers need to assemble, investigate, and run applets and applications written in the Java language.

Android SDK

Android is a working framework taking into account the Linux piece, and composed principally for touchscreen cell phones, for example, cell phones and tablet PCs. At first grew by Android, Inc., which Google supported fiscally and later purchased in 2005, Android was disclosed in 2007 alongside the establishing of the Alliance consortium of equipment, programming, and telecom organizations.

The main hardware platform for Android is the 32-bit ARMv7 construction modeling. The Android-x86 undertaking gives backing to the x86 structural engineering, and Google TV utilizes a unique x86 adaptation of Android. In 2012, Intel processors started to show up on more standard Android stages, for example, telephones. In 2013, Free scale declared backing for Android on its i.MX processor, particularly the i.MX5X and i.MX6X arrangement.

SQLite

SQLite is a relational database administration framework contained in a C programming library. Rather than other database administration frameworks, SQLite is not a different procedure that is gotten to from the customer application, however an essential piece of it

SQLite is ACID-consistent and executes the vast majority of the SQL standard, utilizing a rapidly and pitifully wrote SQL grammar that does not ensure the area uprightness.

SQLite is a prevalent decision as inserted database for neighborhood/customer stockpiling in application programming, for example, web programs. It is ostensibly the most generally sent database motor, as it is utilized today by a few far reaching programs, working frameworks, and implanted frameworks, among others. SQLite has numerous ties to programming languages. The source code for SQLite is in the general public area.

CHAPTER 3: REQUIREMENT ANALYSIS

Requirement Analysis is the first technical step in the software engineering process. It is at this point that a general statement of software scope is refined into a concrete specification that becomes a foundation for all the software engineering activities that follow.

Analysis must focus on the informational, functional, and behavioral domains of a problem. To better understand what is required, models are created, the problem is partitioned, and representation that depict the essence of requirements and later, implementation detail, are developed.

3.1 REQUIREMENT DETERMINATION

An information system is intended to meet the diverse needs of an organization.

The customer relationship management system and sale automation system is required to contain the following features:

- It should contain all the information about the customer such as name, address, salary, occupation, date of birth, phone number.
- It should maintain product details such as product name, features, price, model number, colors.
- It should maintain salesman data such as salesman name, address, salary, occupation, date of birth, phone number, date of joining, manager name.
- It should also maintain the quota of each salesman.
- It should maintain the transaction of products such as sale of the products, addition of the product in the organization etc.
- It should be able to notify that the product stock is about to finish and reorder is required.
- It should check the performance of the salesman by comparing it with some standards set.
- It should feature the security check so that only the authorized person is allowed to manipulate database as per the level assigned.
- It should be user-friendly software.

Requirement analysis task is a process of discovery, refinement, modeling and specification. Both the developers and customers take an active role in requirement analysis. Requirement analysis is a communication intensive activity. Requirement analysis can be divided into:

- Problem Recognition
- Problem Evaluation & Synthesis

3.2 Problem Recognition

The goal of this step is recognition of basic problem elements as indicated by the customer. The basic purpose of this activity is to obtain a thorough understanding of the needs of the client and the user, what exactly is desired from the software and what are the constraints on the solution.

Problems of the existing system:

- Security can’t assured
- Delay in storing and retrieving information
- Possibility of human errors

3.3 Problem Evaluation & Synthesis

In this step analyst must define all externally observable objects, evaluate flow and control of the information, define and elaborate all software functions, understand. Software behavior and design constraints etc. Evaluation and synthesis continues until both analyst and customer field confident about the product.

Once the problems are identified, evaluation process begins. After evaluation of the current problem and desired in formations, the analyst synthesis one or more solutions.

- ❖ Security can be assured
- ❖ Cost effective
- ❖ No chance of errors

3.4 MODELING

During the evaluation and solution synthesis activity, the analyst creates models of the system in an effort to better understand data and control flow. The model serves as foundation for software design and as the basis for the creation of specification for the software. For the better understanding of data and control flow we use Data Flow Diagram.

3.5 Data Flow Diagrams

Data Flow Diagram is used to define the flow of the system and its resources such as information's. DFDs are a way of expressing systems requirements in graphical manner. DFD represents one of the most ingenious tools used for structured analysis. It has the purpose of clarifying system requirements and identifying major transformations that will become programs in the system design. It is the major starting point in the design phase that functionalities decompose the requirement specification to the lowest level of detail.

A data flow diagram is graphical tool used to describe and analyze movement of data through a system. These are the central tool and the basis from which the other components are developed. The transformation of data from input to output, through processed, may be described logically and independently of physical components associated with the system. These are known as the logical data flow diagrams. The physical data flow diagrams show the actual implements and movement of data between people, departments and workstations. A full description of a system actually consists of a set of data flow diagrams. Using two familiar notations Yourdon, Gane and Sarson notation develops the data flow diagrams. Each component in a DFD is labeled with a descriptive name. Process is further identified with a number that will be used for identification purpose. The development of DFD'S is done in several levels. Each process in lower level diagrams can be broken down into a more detailed DFD in the next level. The lop-level diagram is often called context diagram. It consists a single process bit, which plays vital role in studying the current system. The process in the context level diagram is exploded into other process at the first level DFD.

The idea behind the explosion of a process into more process is that understanding at one level of detail is exploded into greater detail at the next level. This is done until further explosion is necessary and an adequate amount of detail is described for analyst to understand the process.

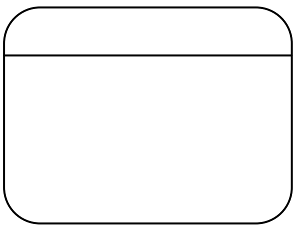
Larry Constantine first developed the DFD as a way of expressing system requirements in a graphical from, this lead to the modular design.

A DFD is also known as a “bubble Chart” has the purpose of clarifying system requirements and identifying major transformations that will become programs in system design. So it is the starting point of the design to the lowest level of detail. A DFD consists of a series of bubbles joined by data flows in the system.

DFD Symbols:

In the DFD, there are four symbols

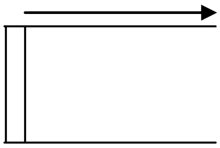
1. A square defines a source(originator) or destination of system data
2. An arrow identifies data flow. It is the pipeline through which the information flows
3. A circle or a bubble represents a process that transforms incoming data flow into outgoing data flows.
4. An open rectangle is a data store, data at rest or a temporary repository of data



Process that transforms data flow.

Source or Destination of data

Data flow



Data Store

Constructing a DFD:

Several rules of thumb are used in drawing DFD’S:

Process should be named and numbered for an easy reference. Each name should Be representative of the process.

The direction of flow is from top to bottom and from left to right. Data traditionally flow from source to the destination although they may flow back to the source. One way to indicate this is to draw long flow line back to a source. An alternative way is to repeat the source symbol as a destination.Since it is used more than once in the DFD it is marked with a short diagonal.

When a process is exploded into lower level details, they are numbered.

The names of data stores and destinations are written in capital letters. Process and dataflow names have the first letter of each work capitalized. A DFD typically shows the minimum contents of data store. Each data store should contain all the data elements that flow in and out.

Questionnaires should contain all the data elements that flow in and out. Missing interfaces

Redundancies and like is then accounted for often through interviews.

Silent Feature of DFD's

1. The DFD shows flow of data, not of control loops and decision are controlled considerations do not appear on a DFD.
2. The DFD does not indicate the time factor involved in any process whether the dataflow take place daily, weekly, monthly or yearly.
3. The sequence of events is not brought out on the DFD.

Data Flow:

1) A Data Flow has only one direction of flow between symbols. It may flow in both directions between a process and a data store to show a read before an update. The later is usually indicated however by two separate arrows since these happen at different type.
2) A join in DFD means that exactly the same data comes from any of two or more different processes data store or sink to a common location.
3) A data flow cannot go directly back to the same process it leads. There must be at least one other process that handles the data flow produce some other data flow returns the original data into the beginning process.
4) A Data flow to a data store means update (delete or change).
5) A data Flow from a data store means retrieve or use. A data flow has a noun phrase label more than one data flow noun phrase can appear on a single arrow as long as all of the flows on the same arrow move together as one package.

Level 1.1 DFD (Admin)

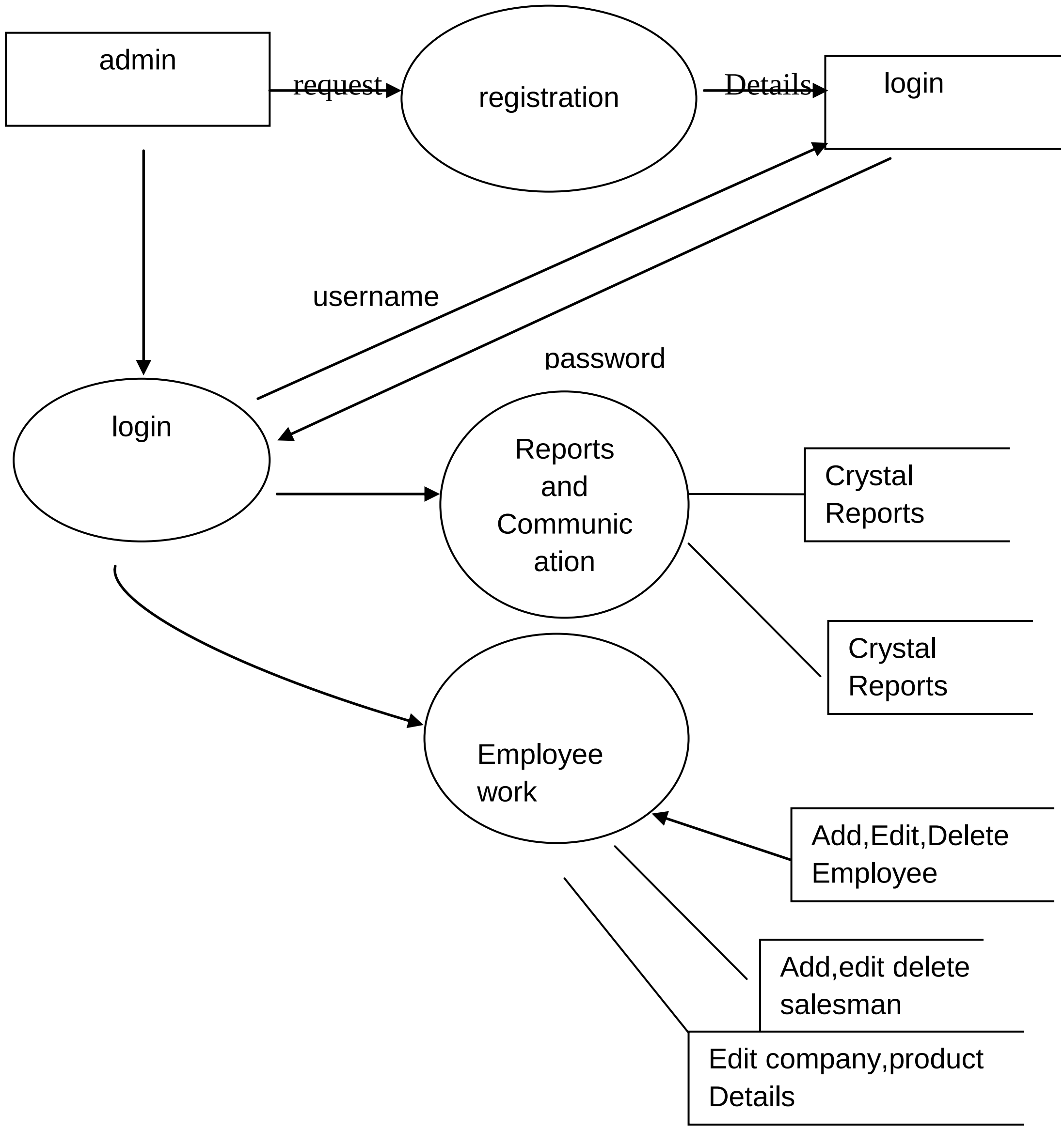


Fig 3.1: Level 1.0 DFD (Admin

Administrator- administrator is responsible for following activities-

- Adding and Modifying Employee Details
- Assign order
- Adding and modifying Product Data
- Adding and modifying Product data
- Performing different analysis on transaction, complaint, sales and other data.

Level 1.2 DFD (Employee)

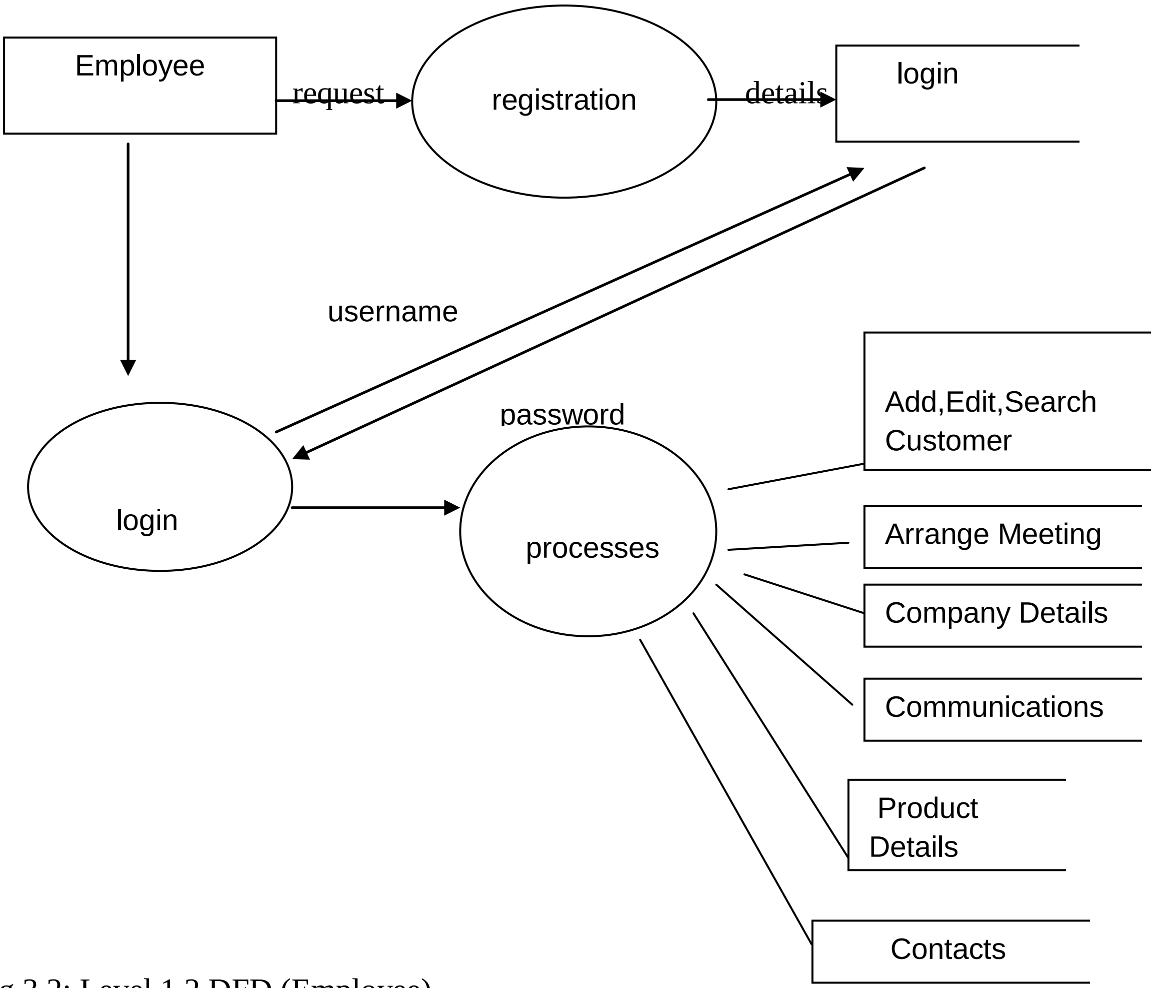


Fig 3.2: Level 1.2 DFD (Employee)

Employee- Salesman is responsible for following activities

- Order processing
- Maintaining customer data
- Sales call management
- Add and communicate with company

Level 1.3 DFD (Customer Support)

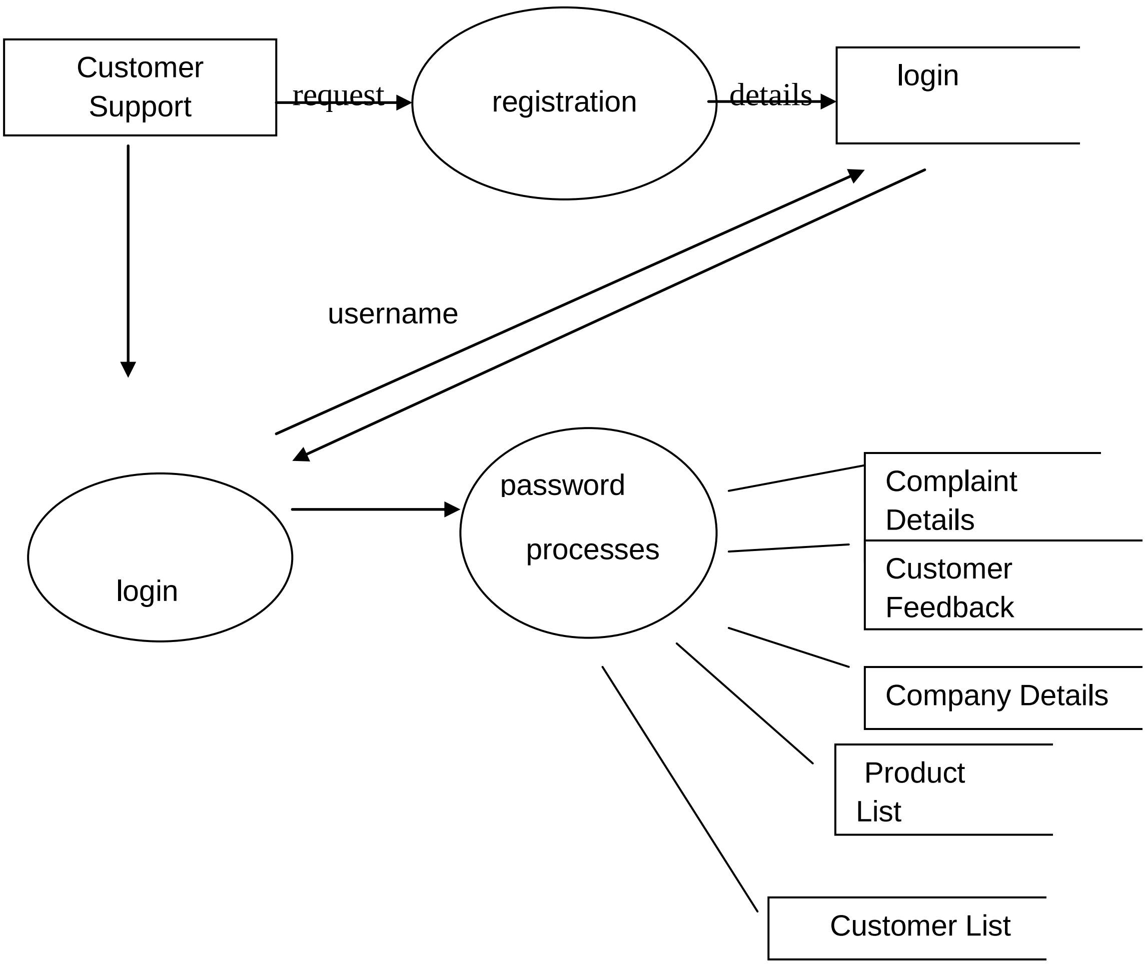


Fig 3.3: Level 1.3 DFD (Customer Support)

Customer Relationship Executive - he is responsible for following activities

- Complaint management
   Input customer feedback

3.6 Flow Chart

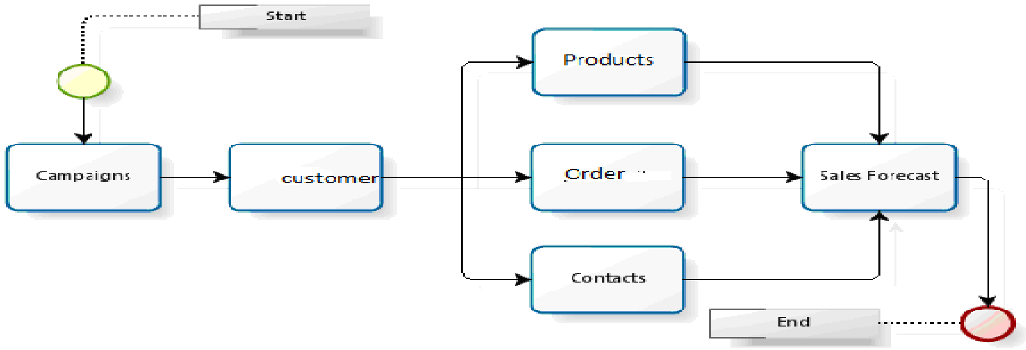


Fig 3.4: Flow Chart

3.7 E-R Diagram

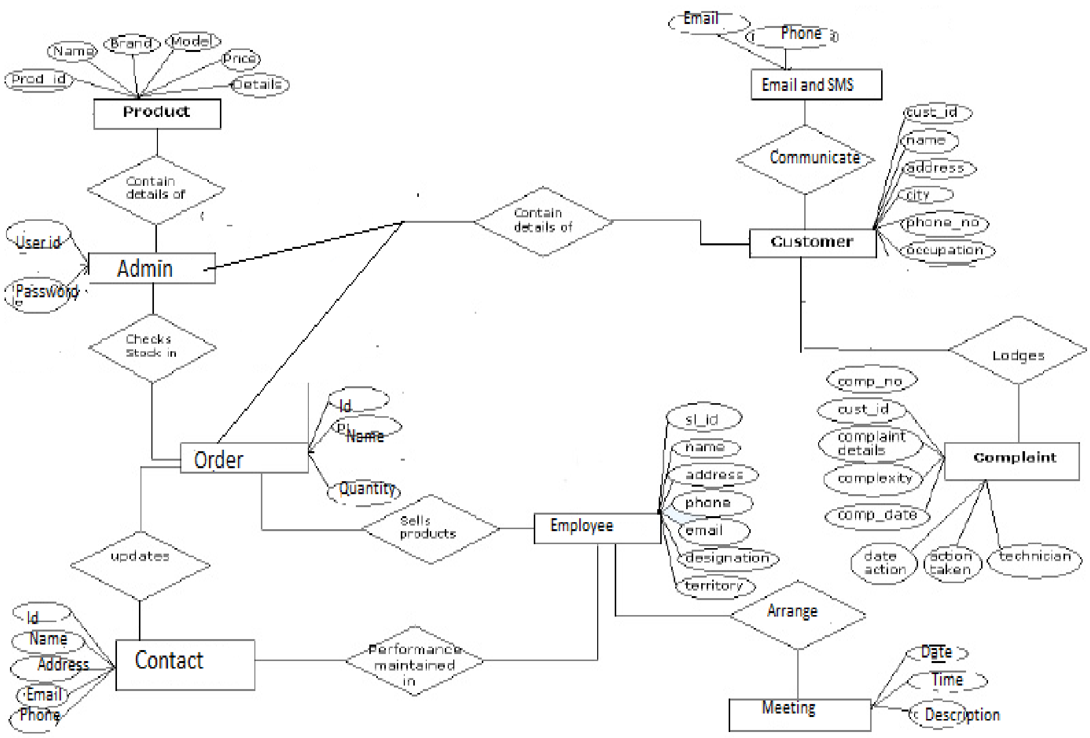


Fig 3.5: E-R Diagram

CHAPTER 4: SOFTWARE REQUIREMENT SPECIFICATIONS

4.1 Introduction

The Software Requirements Specification is produced at the culmination of the analysis task. The function and performance allocated to software as part of System Engineering are refined by:

- Establishing a complete information description of the System.
- A detailed functional description.
- A representation of System behavior .
- An indication of performance requirements and design constraints.
- Appropriate validation criteria, &
- Other information pertinent to requirements.

Purpose

The purpose of the project is to develop a system, which will helpful in management of customer, sales, and complaint data effectively to manage relationship with customer effectively. Proper management of sales data can be used for performing various type of analysis, which can be used for planning marketing strategies and sales planning in effective way.

4.2 Product Scope

The software is accessible by different employees of the organization with different access to data. The data should be very secure because as organization is developing this system to manage good customer relation, this data is high confidential in development of business strategy and marketing plans. Data of customer, product and salesman once enter in the database cannot be deleted as other tables are related to them. Complaint should be resolve as per date of complaint and their priority. Performance Evaluation of marketing strategy should be done time-to-time.

Overall Description

Product Functions

- The organization has many salesmen for selling different products.
- Territory is assigned to each salesman and salesman has to sell product to match quota assign to him.

- After each transaction salesman should maintain data of order and customer.
- There is a list of prospective customer, on the basis of that list salesman can choose customers is their territory to plan their tour in effective way.
- The system should enter data of different complaints of customers and give a complaint number to customer. Pending complaint should to display by the system time to time so that they can be sort out on time.
- Inventory should be checked before taking any order and after processing of order inventory should be updated.
- There should be provision to maintain feedback, query of customers.
- System should perform various analysis on customer, sales and complaint data on the basis of product, territory etc.

User Classes and Characteristics

User classes of the system are as following

2. **Administrator**- administrator is responsible for following activities-
   - Adding and Modifying Employee Details
   - Assign order
   - Adding and modifying Product Data
   - Adding and modifying Product data
   - Performing different analysis on transaction, complaint, sales and other data.
   - Generating different reports
   - Making user account and assigning authorities to them.
3. **Employee**- Salesman is responsible for following activities
   - Order processing
   - Maintaining customer data
   - Sales call management
   - Add and communicate with company
3. **Customer Relationship Executive** - he is responsible for following activities
   - Complaint management
   - Input customer feedback

Operating Environment

The system developed in Android as front end and Sqlite as Back end. Crystel 10 Reporting Engine is also required.
Minimum hardware requirements are Operating system should be Windows Xp. Laser printer should be attached to the system.

4.3 External Interface Requirements:-

User Interfaces
The System should be Graphical User Interface.

Hardware Interfaces
The system will run on System with Pentium 4 processor, 512 MB RAM, 512 MB Cache and 20 GB hard Disk. It will be connected to Laser Printer.

Software Interfaces
The system will run on Window XP Professional. It is based on Android and Sqlite so Sqlite Browser is also required.

4.4 FUNCTIONAL REQUIREMENTS

Data should be given in a correct form in order to avoid getting erroneous results. The value of transportation cost and the assignment cost should be known prior to calculate the optimal transportation and assignment cost.

- In order to prepare the budgets the proper and the correct data is to be considered.
- The end users should input correct figures so that the Ratio Analysis may result as per expectations.
- Data should be checked for validity
- Consistency between the information provided in different forms under a scheme should be checked
- Messages should be given for improper input data, and invalid data item should be ignored.

4.5 PERFORMANCE REQUIREMENTS

The following characteristics were taken care of in developing the system.

1. USER FRIENDLINESS: The system is easy to learn and understand. A naive user can also use the system effectively, without any difficulty.

2. **RESPONSE TIME:** Response time of all the operations is kept low. This has been made possible by careful planning.

3. **ERROR HANDLING:** Response to user errors and undesired situations have been taken care of to ensure that system operates without halting in case of such situation and proper error messages are given to user. Different validations are on different field so that only correct data will be entering in the database.

4. **SAFETY:** The system is able to avoid catastrophic behavior.

5. **ROBUSTNESS:** The system recovers from the undesired events without human intervention.

6. **SECURITY:** The system provides protection of information through the mechanism of password incorporated in it. Therefore only authorized people can access it database. There are three category of user in the system administrator, salesman and customer service executive. All have different authorities of using the system.

7. **ACCURACY:** The system is highly accurate thus its utility is very high.

8. **COST ELEMENT:** Servicing a given demand in the system doesn’t require a lot of money.

ACCEPTANCE CRITERIA

Before accepting the system, we will demonstrate the system work on existing schemes (designed manually) with actual data as well as we will show through test cases that all the conditions are satisfied. The following acceptance criteria were established for the evaluation of the new system.

4.6 HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirement	**(minimum): -**	
Processor	:	Pentium IV or above
Hard Disk	:	20 GB
RAM	:	512 MB
Monitor	:	Color VGA
Printer	:	Laser

Software Requirement (minimum): -

Platform / Operating System	:	Windows Xp and above
Front End. / GUI Tools	**:**	**Eclipse**
RDBMS	**:**	**SQLITE**
Reporting Tool	**:**	**Crystal Report 10**

CHAPTER 5: SYSTEM DESIGN

Design is the most important part of the development phase for any product or system because design is the place where quality is fostered. Design is the only thing, which accurately translates a customer's requirement in to a finished software product or system. The design step produces a data design, an architectural design, and Interface Design and a Procedural Design.
System specialists often refer to this stage as logical description, in contrast to the process of developing the program software, which is referred to as physical design. The system design describes the data to be input, calculated or stored. Individual data items and calculation procedures are written in detail. The procedures tell how to process the data and produce the output.

5.1 Introduction

Software design sits at the technical kernel of the software engineering process and is applied regardless of the development paradigm and area of application. Design is the first step in the development phase for any engineered product or system. The designer's goal is to produce a model or representation of an entity that will later be built. Beginning, once system requirement have been specified and analyzed, system design is the first of the three technical activities -design, code and test that is required to build and verify software.

The importance can be stated with a single word "Quality". Design is the place where quality is fostered in software development. Design provides us with representations of software that can assess for quality. Design is the only way that we can accurately translate a customer's view into a finished software product or system. Software design serves as a foundation for all the software engineering steps that follow. Without a strong design we risk building an unstable system – one that will be difficult to test, one whose quality cannot be assessed until the last stage.

During design, progressive refinement of data structure, program structure, and procedural details are developed reviewed and documented. System design can be viewed from either technical or project management perspective. From the technical point of view, design is comprised of four activities – architectural design, data structure design, interface design and procedural design.

The most creative and challenging phase of the system life cycle is system design. The term design describes a final system and the process by which it is developed. It refers to the technical specifications that will be applied in implementing the proposed system. It also includes the construction of program and designing of output, input, menu, code, database and process of the system.

System output may be report, document or a message. In on-line applications, information is displayed on the screen. The layout sheet for displayed output is similar to the layout chart used for designing output.

On-line data entry makes use of processor that accepts commands and data from the operator through a keyboard or a device such as touch screen or voice input.

Designing the code depends on the programming language chosen and mostly they are not specified while outlining the design of the system. The goal of coding is to translate the design of the system in to code in a programming language. The aim of the code design is to implement the system in best possible manner.

5.2. Data Design

Database design is used to define and specify the structure of objects used in the system. A wide array of design information must be developed during the database design. A database is the collection of interrelated data stored with minimum redundancy to serve many users quickly and efficiently. The general objective of data base design is to make information access easy, quick, inexpensive, and flexible for the user.

5.3. Architehural Design

Program Structure

The product entitled “Customer Relationship Management System” follows bottom up program structure.

Description about bottom up approach:-

In a bottom-up approach the individual base elements of the system are first specified in great detail. These elements are then linked together to form larger subsystems, which then in turn are linked, sometimes in many levels, until a complete top-level system is formed. This strategy often resembles a "seed" model, whereby the beginnings are small, but eventually grow in complexity and completeness. However, "organic strategies", may result in a tangle of elements and subsystems, developed in isolation, and subject to local optimization as opposed to meeting a global purpose

5.4. Procedural Design

Procedural design occurs after data, architectural, and interface designs must be translated in to operational software. The procedural design for each component, represented in graphical, tabular or text based notation, is the primary work product produced during component level design.

Begin

If ((User name is valid)) and corresponding password is correct) then

The user is an authorized person & can access the system

Else

Display invalid user

End If

End

5.5. Interface Design

External machine interface: - We need a high quality modem to connect to the internet. NIC i.e. network interface card is also required to access the internet.TCP/IP and http are the protocols used. If we want some hard copy of any receipts, we need a printer for that.

External system interface: - Since Crime Reporting System requires a database and is working as online, the client machines require proper connection with the server machine.

Human interface: - In project work entitled “Crime Reporting System” the user will interacts with product through Graphical User Interface (GUI) which will be developed in front pages. Since GUIs are the interface, workings with GUIs are very simple and not at all complicated. So any user who had no knowledge about softwares and computers can use this “Crime Reporting System” very simply.

CHAPTER 6: Implementation and coding

Activity_meeting_new_record.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:id="@+id/orders_edit_form_container"
    android:padding="8dp" >
    <LinearLayout
        style="@android:style/ButtonBar"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginBottom="5dp">
        <Button style=""
            android:text="@string/save_button"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:onClick="saveRecordToDb"/>
        <Button style=""
            android:text="@string/cancel_button"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:onClick="cancel"/>

    </LinearLayout>

      <EditText style="@style/FormInputs"
          android:id="@+id/order_new_record_title"
          android:hint="@string/order_title_hint"
          />
      <LinearLayout
          android:layout_width="match_parent"
          android:layout_height="wrap_content"
          android:orientation="horizontal">
          <Button android:id="@+id/orders_order_date_btn"
          android:onClick="openDatePicker"
          android:layout_width="wrap_content"
          android:layout_height="40dp"
          android:layout_gravity="center_horizontal"
          android:text="@string/order_date_hint"
          />
      <EditText style="@style/FormInputs"
          android:layout_width="0dp"
          android:layout_weight="1"
          android:id="@+id/orders_new_record_order_date_field"
          android:background="@android:color/transparent"
          android:textColor="@android:color/white"
          android:layout_marginTop="5dp"
```

```
            android:layout_marginBottom="10dp"
            android:layout_gravity="center"
            android:textAlignment="center"
            android:enabled="false"
            />
        </LinearLayout>
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">
        </LinearLayout>
        <EditText style="@style/FormInputs"
            android:id="@+id/order_new_record_address"
            android:hint="@string/order_address_hint"
            />
      <ExpandableListView android:id="@+id/members_list"
          android:layout_width="match_parent"
          android:layout_height="wrap_content"
          >
        </ExpandableListView>



</LinearLayout>
```

Activity_view_db.xml

```
<android.support.v4.widget.DrawerLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/drawer_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    xmlns:tools="http://schemas.android.com/tools"
    tools:context="net.gesher.minicrm.ViewDbActivity">
<LinearLayout
    android:padding="8dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="@drawable/crmn"
    android:id="@+id/view_db_list_item"
     >

    <AutoCompleteTextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:visibility="invisible"
        android:id="@+id/search_box"
        android:background="@android:color/transparent"
        android:layout_marginBottom="5dp"/>
    <ListView
        android:id="@android:id/list"
        style="@android:style/Widget.Holo.ListView"
        android:layout_height="wrap_content"
        android:layout_width="match_parent"
        android:divider="#eaf9af"
        android:dividerHeight="5dp"
        />
    <TextView android:id="@android:id/empty"
          android:layout_height="wrap_content"
          android:layout_width="match_parent"
          android:text="@string/empty_list_suggestion"
```

```
            style="@style/H1"/>


    </LinearLayout>
    <ListView android:id="@+id/drawer_menu"
            android:layout_width="240dp"
            android:layout_height="match_parent"
            android:layout_gravity="start"
            android:choiceMode="singleChoice"
            android:divider="@android:color/white"
            android:dividerHeight="2dp"
            android:background="#0daa18"/>
</android.support.v4.widget.DrawerLayout>
```

Add_member_dialoge.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="12dp" >

    <Spinner
        android:id="@+id/add_component_spinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:prompt="@string/add_member_spinner_prompt" />

    <EditText
        android:id="@+id/add_member_amount"
        style="@style/FormInputs"
        android:hint="@string/products_amount_hint"
        android:visibility="gone" />

</LinearLayout>
```

Add_member_dropdown_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<!--
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="5dp"
    android:orientation="vertical" >
    <LinearLayout android:orientation="horizontal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">
        <TextView android:id="@+id/titleText1"
            style="@style/ListItemHeader"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
        <TextView android:id="@+id/titleText2"
            style="@style/ListItemHeader"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
    </LinearLayout>
    <TextView android:id="@+id/subtitle"
        style="@style/ListItemSubheader"
        android:layout_width="match_parent"
```

```
            android:layout_height="wrap_content"/>

    </LinearLayout>
    -->


    <CheckedTextView
    xmlns:android="http://schemas.android.com/apk/res/android"
        android:id="@android:id/text1"
        style="?android:attr/spinnerDropDownItemStyle"
        android:singleLine="true"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ellipsize="marquee"
        android:textSize="22sp"
        android:padding="6dp"
        android:textAlignment="inherit"/>
```

add_member_spinner_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TextView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@android:id/text1"
    style="?android:attr/spinnerItemStyle"
    android:singleLine="true"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="22sp"
    android:ellipsize="marquee"
    android:textAlignment="inherit"/>
```

company_display.xml

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp" >
    <TextView style="@style/DisplayLabel"
        android:text="@string/products_title_hint"/>
    <View style="@style/minorHr"/>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <TextView
          android:id="@+id/products_content_title"
          style="@style/FormInputs"
          android:layout_width="0dp"
          android:layout_weight="1"
          />
        <ImageView style="@style/BtnEditLine"/>
    </LinearLayout>

    <TextView style="@style/DisplayLabel"
        android:text="@string/products_subtitle_hint"/>
    <View style="@style/minorHr"/>
    <LinearLayout
```

```xml
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <TextView
          android:id="@+id/products_content_subtitle"
          style="@style/FormInputs"
          android:layout_width="0dp"
          android:layout_weight="1"
          />
        <ImageView style="@style/BtnEditLine"/>
    </LinearLayout>

    <TextView style="@style/DisplayLabel"
        android:text="@string/products_unit_price_hint"/>
    <View style="@style/minorHr"/>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <TextView
          android:id="@+id/products_content_unit_price"
          style="@style/FormInputs"
          android:layout_width="0dp"
          android:layout_weight="1"
          />
        <ImageView style="@style/BtnEditLine"/>
    </LinearLayout>
    <View style="@style/minorHr"/>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
    </LinearLayout>

    <LinearLayout android:id="@+id/products_amount_container"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:visibility="gone">
    <TextView style="@style/DisplayLabel"
        android:text="@string/products_amount_hint"/>
    <View style="@style/minorHr"/>
    <RelativeLayout
        android:layout_height="wrap_content"
        android:layout_width="match_parent">
    <TextView style="@style/FormInputs"
          android:id="@+id/products_display_amount"
          android:layout_width="wrap_content"
          android:layout_alignParentLeft="true"
          android:layout_alignParentStart="true"
          android:paddingLeft="50dp"
          android:paddingStart="50dp"
          />
    <ImageView style="@style/BtnEditLine"
        android:layout_alignParentRight="true"
        android:layout_alignParentEnd="true" />
    </RelativeLayout>
    </LinearLayout>

    <TextView style="@style/DisplayLabel"
        android:text="@string/products_notes_hint"/>
```

```
    <View style="@style/minorHr"/>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <TextView
          android:id="@+id/products_content_notes"
          style="@style/FormInputs"
          android:layout_width="0dp"
          android:layout_weight="1"
          android:inputType="textMultiLine"
          />
        <ImageView style="@style/BtnEditLine"
           />
    </LinearLayout>

</LinearLayout>
</ScrollView>
```

Company_input.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp">

    <TextView style="@style/DisplayLabel"
        android:text="@string/products_title_hint"/>
    <View style="@style/minorHr"/>
      <EditText style="@style/FormInputs"
          android:id="@+id/products_input_title"
          />
      <TextView style="@style/DisplayLabel"
        android:text="@string/products_subtitle_hint"/>
    <View style="@style/minorHr"/>
    <EditText style="@style/FormInputs"
          android:id="@+id/products_input_subtitle"
          />
      <TextView style="@style/DisplayLabel"
        android:text="@string/products_unit_price_hint"/>
    <View style="@style/minorHr"/>
    <EditText style="@style/FormInputs"
          android:id="@+id/products_input_price"
          android:inputType="numberDecimal"
          />
    <View style="@style/minorHr"/>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
    <EditText android:id="@+id/products_input_new_sale_unit"
        style="@style/FormInputs"
        android:layout_width="0dp"
        android:layout_weight="1"
        android:visibility="gone"/>
    </LinearLayout>
```

```
    <LinearLayout android:id="@+id/products_amount_container"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:visibility="gone">
    <TextView style="@style/DisplayLabel"
        android:text="@string/products_amount_hint"/>
    <View style="@style/minorHr"/>
    <EditText style="@style/FormInputs"
          android:id="@+id/products_input_amount"
            android:inputType="number"
          />
    </LinearLayout>

      <TextView style="@style/DisplayLabel"
        android:text="@string/products_notes_hint"/>
    <View style="@style/minorHr"/>
    <EditText style="@style/FormInputs"
          android:id="@+id/products_input_notes"
            android:inputType="textMultiLine"
          />

    <LinearLayout
        style="@android:style/ButtonBar"
        android:id="@+id/products_edit_form_button_bar"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button android:text="@string/save_button"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:onClick="saveRecordToDb"/>
        <Button android:text="@string/cancel_button"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:onClick="cancel"/>

    </LinearLayout>


</LinearLayout>
</ScrollView>
```

Customer_display_list.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
      xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp" >
    <TextView style="@style/DisplayLabel"
        android:text="@string/customer_first_name_hint"/>
    <View style="@style/minorHr"/>
    <TextView
          android:id="@+id/customers_first_name"
          style="@style/FormInputs"
```

```
        />
    <TextView style="@style/DisplayLabel"
        android:text="@string/customer_last_name_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/FormInputs"
          android:id="@+id/customers_last_name"
          />
    <TextView style="@style/DisplayLabel"
        android:text="@string/customer_phone1_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/FormInputs"
          android:id="@+id/customers_phone1"

          />
    <TextView style="@style/DisplayLabel"
        android:text="@string/customer_phone2_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/FormInputs"
          android:id="@+id/customers_phone2"
          />
    <TextView style="@style/DisplayLabel"
        android:text="@string/customer_email_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/FormInputs"
          android:id="@+id/customers_email"
          />
    <TextView style="@style/DisplayLabel"
        android:text="@string/customer_address_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/FormInputs"
          android:id="@+id/customers_address"
          />
    <TextView style="@style/DisplayLabel"
        android:text="@string/customer_contact_name_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/FormInputs"
          android:id="@+id/customers_contact_name"
          />
    <TextView style="@style/DisplayLabel"
        android:text="@string/customer_contact_phone_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/FormInputs"
          android:id="@+id/customers_contact_phone"
          />
</LinearLayout>
```

Layout_customer_input.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp" >
    <TextView style="@style/DisplayLabel"
        android:text="@string/customer_first_name_hint"/>
```

```
<View style="@style/minorHr"/>
<EditText style="@style/FormInputs"
      android:id="@+id/customers_input_first_name"
      android:hint="@string/customer_first_name_hint"
      />
<TextView style="@style/DisplayLabel"
    android:text="@string/customer_last_name_hint"/>
<View style="@style/minorHr"/>
<EditText style="@style/FormInputs"
      android:id="@+id/customers_input_last_name"
      android:hint="@string/customer_last_name_hint"
      />
<TextView style="@style/DisplayLabel"
    android:text="@string/customer_phone1_hint"/>
<View style="@style/minorHr"/>
<EditText style="@style/FormInputs"
      android:id="@+id/customers_input_phone1"
      android:hint="@string/customer_phone1_hint"
      android:inputType="phone"
      />
<TextView style="@style/DisplayLabel"
    android:text="@string/customer_phone2_hint"/>
<View style="@style/minorHr"/>
<EditText style="@style/FormInputs"
      android:id="@+id/customers_input_phone2"
      android:hint="@string/customer_phone2_hint"
      android:inputType="phone"
      />
<TextView style="@style/DisplayLabel"
    android:text="@string/customer_email_hint"/>
<View style="@style/minorHr"/>
<EditText style="@style/FormInputs"
      android:id="@+id/customers_input_email"
      android:hint="@string/customer_email_hint"
      android:inputType="textEmailAddress"
      />
<TextView style="@style/DisplayLabel"
    android:text="@string/customer_address_hint"/>
<View style="@style/minorHr"/>
<EditText style="@style/FormInputs"
      android:id="@+id/customers_input_address"
      android:hint="@string/customer_address_hint"
      />
<TextView style="@style/DisplayLabel"
    android:text="@string/customer_contact_name_hint"/>
<View style="@style/minorHr"/>
<EditText style="@style/FormInputs"
      android:id="@+id/customers_input_contact_name"
      android:hint="@string/customer_contact_name_hint"
      android:inputType="textPersonName"
      />
<TextView style="@style/DisplayLabel"
    android:text="@string/customer_contact_phone_hint"/>
<View style="@style/minorHr"/>
<EditText style="@style/FormInputs"
      android:id="@+id/customers_input_contact_phone"
      android:hint="@string/customer_contact_phone_hint"
      android:inputType="phone"
      />
<LinearLayout
    style="@android:style/ButtonBar"
```

```xml
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:id="@+id/customers_edit_form_button_bar"
        >
        <Button style=""
            android:text="@string/save_button"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:onClick="saveRecordToDb"/>
        <Button style=""
            android:text="@string/cancel_button"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:onClick="cancel"/>

    </LinearLayout>

</LinearLayout>
</ScrollView>
```

Listview_item.xml

```xml
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    style="@android:style/Widget.ListView.White"
    android:orientation="vertical"
    android:id="@+id/orders_member_list_item_3"
    android:layout_height="wrap_content"
    android:layout_width="match_parent"
    android:padding="3dp"
    android:background="@color/list_group_child_background" >

    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content">
    <LinearLayout android:layout_height="wrap_content"
android:layout_width="match_parent"
        android:orientation="horizontal"
        android:paddingEnd="30dp"
        android:paddingRight="30dp"
        android:layout_alignParentStart="true"
        android:layout_alignParentLeft="true">
          <TextView style="@style/ListItemHeader"
              android:id="@+id/field_title1"
              android:layout_height="wrap_content"
              android:layout_width="wrap_content"
              android:paddingEnd="4dp"
              android:paddingRight="4dp"
              />
          <TextView style="@style/ListItemHeader"
              android:id="@+id/field_title2"
              android:layout_height="wrap_content"
              android:layout_width="wrap_content"
             android:paddingEnd="4dp"
             android:paddingRight="4dp"
              />
```

```xml
    </LinearLayout>
    <TextView android:id="@+id/products_display_amount"
              android:layout_height="match_parent"
              android:layout_width="wrap_content"
              android:layout_margin="3dp"
              android:layout_alignParentRight="true"
              android:layout_alignParentEnd="true"
              android:textSize="26sp"
              android:textColor="@android:color/holo_blue_dark"/>
    </RelativeLayout>
      <TextView
        style="@style/ListItemSubheader"
        android:id="@+id/field_subtitle1"
        android:layout_height="wrap_content"
        android:layout_width="match_parent"

        />
      <TextView
        style="@style/ListItemSubheader"
        android:id="@+id/field_subtitle2"
        android:layout_height="wrap_content"
        android:layout_width="match_parent"

        />

</LinearLayout>
```

Login.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
    android:layout_height="match_parent"
android:gravity="center_vertical|center_horizontal"
    android:orientation="vertical"
android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    android:background="@drawable/crmn"
    android:descendantFocusability="beforeDescendants"
    android:focusableInTouchMode="true">

    <LinearLayout
        android:id="@+id/ll1"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:gravity="center_vertical|center_horizontal">

        <TextView
            android:id="@+id/textView2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="ANDROID CRM"

android:textAppearance="?android:attr/textAppearanceLarge"
            android:textColor="#514629"
            android:textSize="30dp" />
```

```
    </LinearLayout>

    <LinearLayout
        android:id="@+id/ll2"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:gravity="center_vertical|center_horizontal">
        <ProgressBar android:id="@+id/login_progress"
style="?android:attr/progressBarStyleLarge"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
            android:visibility="gone" />
        <ScrollView android:id="@+id/login_form"
android:layout_width="match_parent"
            android:layout_height="wrap_content">
            <LinearLayout android:layout_width="match_parent"
android:layout_height="wrap_content"
                android:orientation="vertical">

                <LinearLayout
                    android:id="@+id/email_login_form"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:orientation="vertical" >

                    <EditText
                        android:id="@+id/email"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:layout_marginBottom="10dp"
                        android:hint="User Id"
                        android:inputType="textEmailAddress"
                        android:maxLines="1"
                        android:singleLine="true"
                        android:textColor="#000000"

android:background="@android:color/transparent"

                        android:textColorHint="#000000" />

                    <EditText
                        android:id="@+id/password"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:layout_marginBottom="10dp"
                        android:hint="Password"
                        android:imeActionLabel="LogIn"
                        android:imeOptions="actionUnspecified"
                        android:inputType="textPassword"
                        android:maxLines="1"
                        android:singleLine="true"

android:background="@android:color/transparent"
                        android:textColor="#000000"
                        android:textColorHint="#000000" />

                    <Button
                        android:id="@+id/email_sign_in_button"
                        style="?android:textAppearanceSmall"
```

```xml
                        android:layout_width="match_parent"
                        android:layout_height="32dp"
                        android:layout_marginTop="16dp"

android:background="@android:color/transparent"
                        android:text="Log In"
                        android:textColor="#ffffffff"
                        android:textStyle="bold"

                        android:onClick="OnLogin"/>
                </LinearLayout>

            </LinearLayout>
        </ScrollView>
    </LinearLayout>

    <LinearLayout
        android:id="@+id/ll3"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:gravity="bottom|center_horizontal">

        <Button
            android:id="@+id/signUpTextView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:background="@android:color/transparent"
            android:text="Registration"
            android:textColor="#ffffffff" />

    </LinearLayout>
</LinearLayout>
```

Main_screen_listview_item1.xml

```xml
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    style="@android:style/Widget.ListView.White"
    android:orientation="vertical"
    android:id="@+id/view_db_orders_list_item"
    tools:context="net.gesher.minicrm.ViewDbActivity"
    android:layout_height="wrap_content"
    android:layout_width="match_parent"
    android:padding="3dp"
    android:background="#ffffff" >
     <!--
    <CheckBox
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/view_activity_list_checkbox"
        android:onClick="selectRecord"
        style="?android:checkboxStyle"
        android:focusable="false"
        /> -->

    <TextView
        style="@style/ListItemHeader"
        android:id="@+id/field_title"
        android:layout_height="wrap_content"
        android:layout_width="match_parent"
```

```
        />
    <LinearLayout android:layout_height="wrap_content"
android:layout_width="match_parent"
android:orientation="horizontal">
    <TextView style="@style/ListItemSubheader"
        android:id="@+id/field_subtitle1"
        android:layout_height="wrap_content"
        android:layout_width="0dp"
        android:layout_weight="2"
        />
    <TextView style="@style/ListItemSubheader"
        android:id="@+id/field_subtitle2"
        android:layout_height="wrap_content"
        android:layout_width="0dp"
        android:layout_weight="2"
        />
    </LinearLayout>

</LinearLayout>
```

Main_screen_display_item2.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    style="@android:style/Widget.ListView.White"
    android:orientation="vertical"
    android:id="@+id/view_db_orders_list_item"
    tools:context="net.gesher.minicrm.ViewDbActivity"
    android:layout_height="wrap_content"
    android:layout_width="match_parent"
    android:padding="3dp"
    android:background="#ffffff" >
     <!--

    <CheckBox
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/view_activity_list_checkbox"
        android:onClick="selectRecord"
        style="?android:checkboxStyle"
        android:focusable="false"
        /> -->
    <LinearLayout android:layout_height="wrap_content"
android:layout_width="match_parent"
android:orientation="horizontal">
            <TextView style="@style/ListItemHeader"
                android:id="@+id/field_title1"
                android:layout_height="wrap_content"
                android:layout_width="wrap_content"
                android:paddingRight="4dp"
                />
            <TextView style="@style/ListItemHeader"
                android:id="@+id/field_title2"
                android:layout_height="wrap_content"
                android:layout_width="wrap_content"
               android:paddingRight="4dp"
                />
    </LinearLayout>
      <TextView
        style="@style/ListItemSubheader"
```

```xml
        android:id="@+id/field_subtitle"
        android:layout_height="wrap_content"
        android:layout_width="match_parent"

        />
</LinearLayout>
```

Splash_layout.xml

```xml
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/crmn"
    android:alpha="125"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin" >

    <ProgressBar
        android:id="@+id/progressBar1"
        style="?android:attr/progressBarStyleLarge"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:layout_centerHorizontal="true"
        android:layout_centerVertical="true"
        android:visibility="invisible" />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignTop="@+id/progressBar1"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="24dp"
        android:background="@android:color/transparent"
        android:text="Log In" />

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/button1"
        android:layout_centerVertical="true"
        android:background="@android:color/transparent"
        android:text="Credit" />

</RelativeLayout>
```

Contacts_input.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
```

```xml
        android:padding="8dp" >

        <TextView style="@style/DisplayLabel"
            android:text="@string/workers_fname_hint"/>
        <View style="@style/minorHr"/>
          <EditText style="@style/FormInputs"
              android:id="@+id/worker_new_record_first_name"
              />
          <TextView style="@style/DisplayLabel"
            android:text="@string/workers_lname_hint"/>
        <View style="@style/minorHr"/>
        <EditText style="@style/FormInputs"
              android:id="@+id/worker_new_record_last_name"
              />
          <TextView style="@style/DisplayLabel"
            android:text="@string/workers_phone1_hint"/>
        <View style="@style/minorHr"/>
        <EditText style="@style/FormInputs"
              android:id="@+id/worker_new_record_phone1"
              android:inputType="phone"
              />
        <TextView style="@style/DisplayLabel"
            android:text="@string/workers_phone2_hint"/>
        <View style="@style/minorHr"/>
        <EditText style="@style/FormInputs"
              android:id="@+id/worker_new_record_phone2"
              android:inputType="phone"
              />
          <TextView style="@style/DisplayLabel"
            android:text="@string/workers_email_hint"/>
        <View style="@style/minorHr"/>
        <EditText style="@style/FormInputs"
              android:id="@+id/worker_new_record_email"
              android:inputType="textEmailAddress"
              />
          <TextView style="@style/DisplayLabel"
            android:text="@string/workers_address_hint"/>
        <View style="@style/minorHr"/>
        <EditText style="@style/FormInputs"
              android:id="@+id/worker_new_record_address"
              />
        <TextView style="@style/DisplayLabel"
            android:text="@string/workers_occupation_hint"/>
        <View style="@style/minorHr"/>
        <EditText style="@style/FormInputs"
              android:id="@+id/worker_new_record_occupation"
              android:hint="@string/workers_occupation_hint"
              />
          <LinearLayout android:layout_width="match_parent"
              android:layout_height="wrap_content"
              android:orientation="horizontal">
              <TextView style="@style/DisplayLabel"
                  android:layout_width="wrap_content"
                  android:layout_weight="0"
            android:text="@string/workers_relationship_hint"/>
              <Spinner android:id="@+id/workers_relationship_spinner"
                  android:layout_height="wrap_content"
                  android:layout_width="0dp"
                  android:layout_weight="1"
                  android:entries="@array/workers_occupations"/>
          </LinearLayout>
```

```
    <LinearLayout
        style="@android:style/ButtonBar"
        android:id="@+id/workers_edit_form_button_bar"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button android:text="@string/save_button"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:onClick="saveRecordToDb"/>
        <Button android:text="@string/cancel_button"
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:onClick="cancel"/>

    </LinearLayout>


</LinearLayout>
</ScrollView>
```

Contacts_display_form.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp" >
    <TextView style="@style/DisplayLabel"
        android:text="@string/workers_fname_hint"/>
    <View style="@style/minorHr"/>
    <TextView
          android:id="@+id/workers_first_name"
          style="@style/DisplayContent"
          />
    <TextView style="@style/DisplayLabel"
        android:text="@string/workers_lname_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/DisplayContent"
          android:id="@+id/workers_last_name"
          />
    <TextView style="@style/DisplayLabel"
        android:text="@string/workers_phone1_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/DisplayContent"
          android:id="@+id/workers_phone1"

          />
    <TextView style="@style/DisplayLabel"
        android:text="@string/workers_phone2_hint"/>
    <View style="@style/minorHr"/>
    <TextView style="@style/DisplayContent"
          android:id="@+id/workers_phone2"
          />
    <TextView style="@style/DisplayLabel"
```

```
            android:text="@string/workers_email_hint"/>
        <View style="@style/minorHr"/>
        <TextView style="@style/DisplayContent"
              android:id="@+id/workers_email"
              />
        <TextView style="@style/DisplayLabel"
            android:text="@string/workers_address_hint"/>
        <View style="@style/minorHr"/>
        <TextView style="@style/DisplayContent"
              android:id="@+id/workers_address"
              />
        <TextView style="@style/DisplayLabel"
            android:text="@string/workers_occupation_hint"/>
        <View style="@style/minorHr"/>
        <TextView style="@style/DisplayContent"
              android:id="@+id/workers_occupation"
              />
        <TextView style="@style/DisplayLabel"
            android:text="@string/workers_relationship_hint"/>
        <View style="@style/minorHr"/>
        <TextView style="@style/DisplayContent"
              android:id="@+id/workers_relationship"
              />
    </LinearLayout>
</ScrollView>
```

6.2. Implementation Methods

The term implementation has different meanings, ranging from the conversion of a basic application to a complete replacement of a computer system. The procedure, however, is virtually the same. Implementation is used here to mean the process for converting a new or revised system into an operational one. Conversion is one aspect of implementation. The other aspects are the post implementation review and software maintenance.

There are three types of implementation:

- ❖ Implementation of a computer system to replace a manual system.
- ❖ Implementation of a new computer system to replace an existing one.
- ❖ Implementation of a modified application to replace an existing one.

Conversion means changing from one system to another. That is data in the old format is run through a program, or a series of programs, to convert it into the new format. Conversion can also be from one hardware medium to another. The objective is to put the tested system into operation while holding costs, risk and personnel irritation to a minimum. It involves :-

- ❖ Creating computer compatible files.
- ❖ Training the operating staff.
- ❖ Installing terminals and hardware.

A problem for management is discovering, at integration time, that pieces of modules simply do not fit together. Such problem will arise when charge is made to only one copy of the design document. When integration is completed, the product as whole is tested, this is termed product testing. When the developers are confident about the correction of every aspect of product, it is handled over to the client for acceptance testing.

Implementation Plan

The implementation plan includes a description of all the activities that must occur to implement the new system and to put it into operation. It identifies the personnel responsible for the activities and prepares a time chart for implementing the system. The implementation plan consists of the following steps.

- List all files required for implementation.
- Identify all data required to build new files during the implementation.
- List all new documents and procedures that go into the new system.

The implementation plan should anticipate possible problems and must be able to deal with them. The usual problems may be missing documents; mixed data formats between current and files, errors in data translation, missing data etc.

CHAPTER 7:SYSTEM TESTING

7.1 Introduction

Software testing is the process used to help identify the correctness, completeness, security, and quality of developed computer software. Testing is a process of executing a program or application with the intent of finding errors. With that in mind, testing can never completely establish the correctness of arbitrary computer software. In other words, testing is criticism or comparison that is comparing the actual value with an expected one. An important point is that software testing should be distinguished from the separate discipline of software quality assurance, which encompasses all business process areas, not just testing.

TESTING OBJECTIVES :

- Testing is a process of executing a program with the intent of finding an error.
- A good test case is one that has a high-probability of finding an as-yet-undiscovered error.
-

A successful test is one that uncovers an as-yet-undiscovered error.

Software Testing Techniques

The importance of software testing and its impact on software cannot be underestimated. Software testing is a fundamental component of software quality assurance and represents a review of specification, design and coding. The greater visibility of software systems and the cost associated with software failure are motivating factors for planning, through testing. It is not uncommon for a software organization to spent 40% of its effort on testing.

7.2 White Box Testing

White box testing is a test case design approach that employs the control architecture of the procedural design to produce test cases. Using white box testing approaches, the software engineering can produce test cases that

(1) Guarantee that all independent paths in a module have been exercised at least once

(2) Exercise all logical decisions\

(3) Execute all loops at their boundaries and in their operational bounds

(4) Exercise internal data structures to maintain their validity.

Various white box techniques

Basis Path Testing

Basic path testing is a white box testing techniques that allows the test case designer to produce a logical complexity measure of procedural design and use this measure as an approach for outlining a basic set of execution paths. Test cases produced to exercise each statement in the program at least one time during testing.

Control Structure Testing

Although basis path testing is simple and highly effective, it is not enough in itself. Next we consider variations on control structure testing that broaden testing coverage and improve the quality of white box testing. Different control structure techniques are

a. Condition testing

b. Data flow testing

c. Loop testing

7.3 Black Box Testing

Black Box Testing is not a type of testing; it instead is a testing strategy, which does not need any knowledge of internal design or code etc. As the name "black box" suggests, no knowledge of internal logic or code structure is required. The types of testing under this strategy are totally based/focused on the testing for requirements and functionality of the work product/software application.

Various black box techniques

Functional Testing:

In this type of testing, the software is tested for the functional requirements. The tests are written in order to check if the application behaves as expected.

Smoke Testing:

This type of testing is also called sanity testing and is done in order to check if the application is ready for further major testing and is working properly without failing up to least expected level.

Recovery Testing:

Recovery testing is basically done in order to check how fast and better the application can recover against any type of crash or hardware failure etc. Type or extent of recovery is specified in the requirement specifications.

Alpha Testing:

In this type of testing, the users are invited at the development center where they use the application and the developers note every particular input or action carried out by the user. Any type of abnormal behavior of the system is noted and rectified by the developers.

Beta Testing:

In this type of testing, the software is distributed as a beta version to the users and users test the application at their sites. As the users explore the software, in case if any exception/defect occurs that is reported to the developers.

7.4 Software Testing Strategies in used in the project

A strategy for software testing integrates software test case design techniques into a well-planned set of steps that cause the production of software. A software test strategy provides a road map for the software developer, the quality assurance organization, and the customer

7.4.1 Unit testing

Unit testing concentrates verification on the smallest element of the program – the module. Using the detailed design description important control paths are tested to establish errors within the bounds of the module.

Firstly the unit testing on various modules and sub modules is performed in the project. Different modules are tested with different correct and incorrect data. For example in the order processing module order of 0 product is not allowed so in this case different methods are used to find out whether the modules is performing all processes correctly. All modules are tested to find out that whether they are working properly.

7.4.2 Integration testing :-

Once all the individual units have been tested there is a need to test how they were put together to ensure no data is lost across interface, one module does not have an adverse impact on another and a function is not performed correctly. Integration testing is a systematic approach that produces the program structure while at the same time producing tests to identify errors associated with interfacing.

In this project bottom up integration testing is used. Firstly lower level modules are tested. As modules are integrated bottom up, processing required for modules subordinates to a given level is always available and the need for stubs is eliminated.

7.4.3 Validation testing :-

As a culmination of testing, software is completely assembled as a package, interfacing errors have been identified and corrected, and a final set of software tests validation testing are started. Validation can be defined in various ways, but a basic one is valid succeeds when the software functions in a fashion that can reasonably expected by the customer.

In the first phase of alpha testing, developers test the software using white box techniques.

Additional inspection is then performed using black box techniques. This is usually done by a dedicated testing team. This is often known as the second stage of alpha testing.

Unit and integrated tests concentrate on functional verification of a component and incorporation of components into a program structure. Validation testing demonstrates tractability to software requirements, and system testing validates software once it has been incorporated into a larger system.

7.5 SYSTEM IMPLEMENTATION

The purpose of **System Implementation** can be summarized as follows: making the new system available to a prepared set of users (the deployment), and positioning on-going support and maintenance of the system within the Performing Organization (the transition). At a finer level of detail, deploying the system consists of executing all steps necessary to educate the Consumers on the use of the new system, placing the newly developed system into production, confirming that all data required at the start of operations is available and accurate, and validating that business functions that interact with the system are functioning properly. Transitioning the system support responsibilities involves changing from a system development to a system support and maintenance mode of operation, with ownership of the new system moving from the Project Team to the Performing Organization.

Once the design model of **"CRM "** is created, it is implemented as a prototype, examined by end -users and modified by developers, i.e. us, based on their comments. To accommodate this iterative design approach, a broad class of interface design and prototyping tools has evolved.

This phase consists of the following processes:

- **Prepare for System Implementation,** where all steps needed in advance of actually deploying the application are performed, including preparation of both the production environment and the Consumer communities.

- **Deploy System,** where the full deployment plan, initially developed during System Design and evolved throughout subsequent lifecycle phases, is executed and validated.

- **Transition to Performing Organization,** where responsibility for and ownership of the application are transitioned from the Project Team to the unit in the Performing Organization that will provide system support and maintenance.

Prepare for system implementation

In the implementation of any new system, it is necessary to ensure that the Consumer community is best positioned to utilize the system once deployment efforts have been validated. Therefore, all necessary training activities must be scheduled and coordinated. As this training is often the first exposure to the system for many individuals, it should be conducted as professionally and competently as possible. A positive training experience is a great first step towards Customer acceptance of the system. Often the performance of deployment efforts impacts many of the Performing Organization's normal business operations. Examples of these impacts include:

- Consumers may experience a period of time in which the systems that they depend on to perform their jobs are temporarily unavailable to them.

- Technical Services personnel may be required to assume significant implementation responsibilities while at the same time having to continue current levels of service on other critical business systems.

- Technical Support personnel may experience unusually high volumes of support requests due to the possible disruption of day-to-day processing.

DEPLOY SYSTEM

Deploying the system is the culmination of all prior efforts where all of the meetings, planning sessions, deliverable reviews, prototypes, development, and testing pay off in the delivery of the final system. It is also the point in the project that often requires the most coordination, due to the breadth and variety of activities that must be performed. Depending upon the complexity of the system being implemented, it may impact technical, operational, and cultural aspects of the organization. A representative sample of high-level activities might include the installation of new hardware, increased network capabilities, deployment and configuration of the new system software, a training and awareness campaign, activation of new job titles and responsibilities, and a completely new operational support structure aimed at providing Consumer-oriented assistance during the hours that the new system is available for use (to name a few).

TRANSITION TO PERFORMING ORGANIZATION

In many organizations, the team of individuals responsible for the long-term support and maintenance of a system is different from the team initially responsible for designing and developing the application. Often, the two teams include a comparable set of technical skills. The responsibilities associated with supporting an operational system, however, are different from those associated with new development. In order to affect this shift of responsibilities, the Project Team must provide those responsible for system support in the Performing Organization with a combination of technical documentation, training, and hands-on assistance to enable them to provide an acceptable level of operational support to the Consumers. This system transition is one element

7.6 SYSTEM MAINTENANCE

Software once designed & developed needs to be maintained till end of software / system life. When the system is fully implemented, analyst must take precautions to ensure that the need for maintenance is controlled through design and testing and the abili ty to perform it is provided through proper design practices.

Maintenance is necessary to eliminate errors in the working system during its working life and to tune the system to any variations in its working environment. Often small system deficiencies a re found as a system is brought into operations and changes are made to remove them. System planners must always plan for resource availability to carry out these maintenance functions. The importance of maintenance is to bring the new system to standards.

The term **"Software Maintenance"** is commonly used to refer to the modification that are made to a software system in its initial release.

Maintenance requirements for information systems

(a) From 60% to 90% of the overall cost of software during the life of a system is spend on maintenance.

(b) Often maintenance is not done very efficiently.

(c) Software demand is growing at a faster rate than supply. Many programmers spending more time on systems maintenance than on new software development.

The keys to reduce the need of maintenance , while making it possible to do essential tasks more efficiently, are as follows :

- More accurately defining the user's requirements during systems development.
- Making better Systems documentation.
- Using proper methods of designing processing logic and communicating it to project team members.
- Utilising the existing tools and techniques in an effective way.
- Managing the systems engineering process in a better and effective way.

7.7 SYSTEM SECURITY MEASURES

The System designed & developed is to be kept secure from various persons or users who are not allowed to use the software. Various Security measures are taken to prevent the Software / System from other persons or users.
In this System, users are divided in three category-

- **Administrator** has authority of maintaining salesman data, product data, assigning territory to the salesman, stock processing, offer letter processing, quota assignment, creating user account. Administrator can also provide facility of analyzing customer, sales, and complaint data.
- **Employee** has authority of adding customer details, order processing, sales call data management.
- **Customer** Support executive has authority of complaint processing and entering customer feedback.

Different authority is assigned different user. Each user has a user name and password to access the system. Every time when user will open the system, user has to give his/her username and password in the login window. Access category is assigned to every user. On the basis of this category user get access in the system.

CHAPTER 8: RESULTS AND SCREENSHOTS

Main Page

Splash Screen

All Registered Customer Information List

New Customer Add form

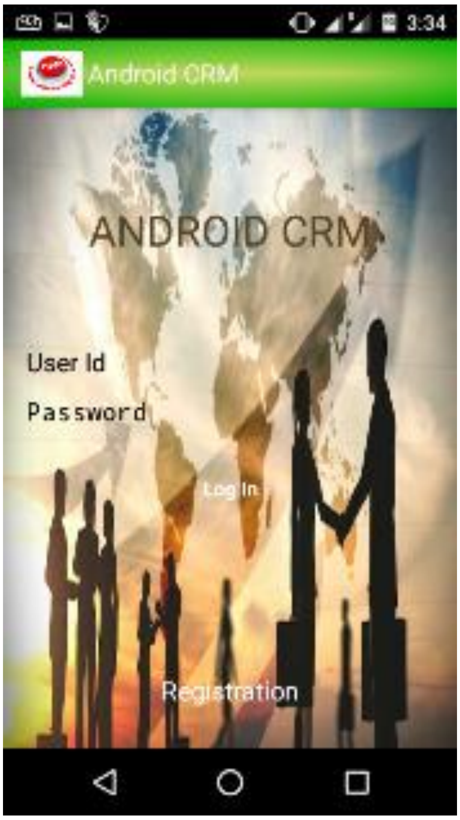
Edit Meeting Information

Show Meeting description of Individual Meeting

Display All meeting List

Add new meetings with customer and company

Employee Login Page



Add Customer,contact with action bar

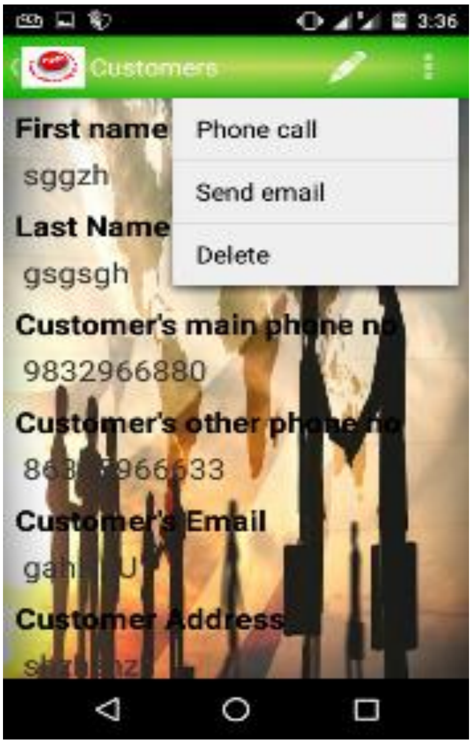
Search and Sorting options for finding customer,contact,meeting details

Phone call and email Sending features for individual customer,company,contact

Add contact

Add new Company

Add new Customers



Edit Customer Details

CHAPTER 9: FUTURE SCOPE OF THE PROJECT

This application can be implement in those organizations where there is need for automation of sales work. This application minimize manual work in the organization so work load on organization will decrease and efficiency and effectiveness of the organization will increase. This application helps in reducing redundancy of the data and provides security so that unauthorized person cannot access the application. The database management in the organization becomes more reliable. This application provide facility to analyze data of customer, sale and complains. This can be highly useful in planning marketing and sales strategy of the organization. It attempts to integrate and automate the various customer-serving processes within a company. CRM is a strategy used to learn more about customers' needs and behaviors in order to develop stronger relationships with them. Good customer relationships are at the heart of business success. So this application will help in building good customer relation with customer.

With the help of this project, the organization will be able to managing customer data in effective way. Proper management of sales and customer data will provide facility of obtaining customer details time to time on the basis of which organization can make marketing and sales plan effectively

FURTHER ENHANCEMENT OF THE PROJECT

There is always room for improving in any software package, however good efficient it may be. But the important thing is that the system should be flexible enough for future modification/ alteration whenever and by whomsoever it may be. Keeping in consideration this important factor, the system is designed in such a way. The software is developed in modules are efficient enough to introduce any change in the software to get more information.

- Similarly, the present system can be implemented on Internet and software can be connected to the various branches of this home appliance company of course with more security constraints added to it.
- This project can be attached to website of the company which may provide information related to products and also provide facility of registering product etc. It may also help in finding new prospective customer.
- The backend can be improved using Oracle; it will provide better database management and securities.

- More modules can be added in the system such as it can provide facility of direct email so that organization can generate offer letter and send it directly to the customer.

CHAPTER 10: REFERENCES

Websites

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